Chapter 7.

Telecommunications, the internet, and wireless technology

Questions to Answer

1. What are the principal components of telecommunications networks and key networking technologies?

2. What are the different types of networks?

3. How do the Internet and Internet technology work, and how do they support communication and e-business?

4. What are the principal technologies and standards for wireless networking, communication, and Internet access?

7.1

What are the principal components of telecommunications networks and key networking technologies?

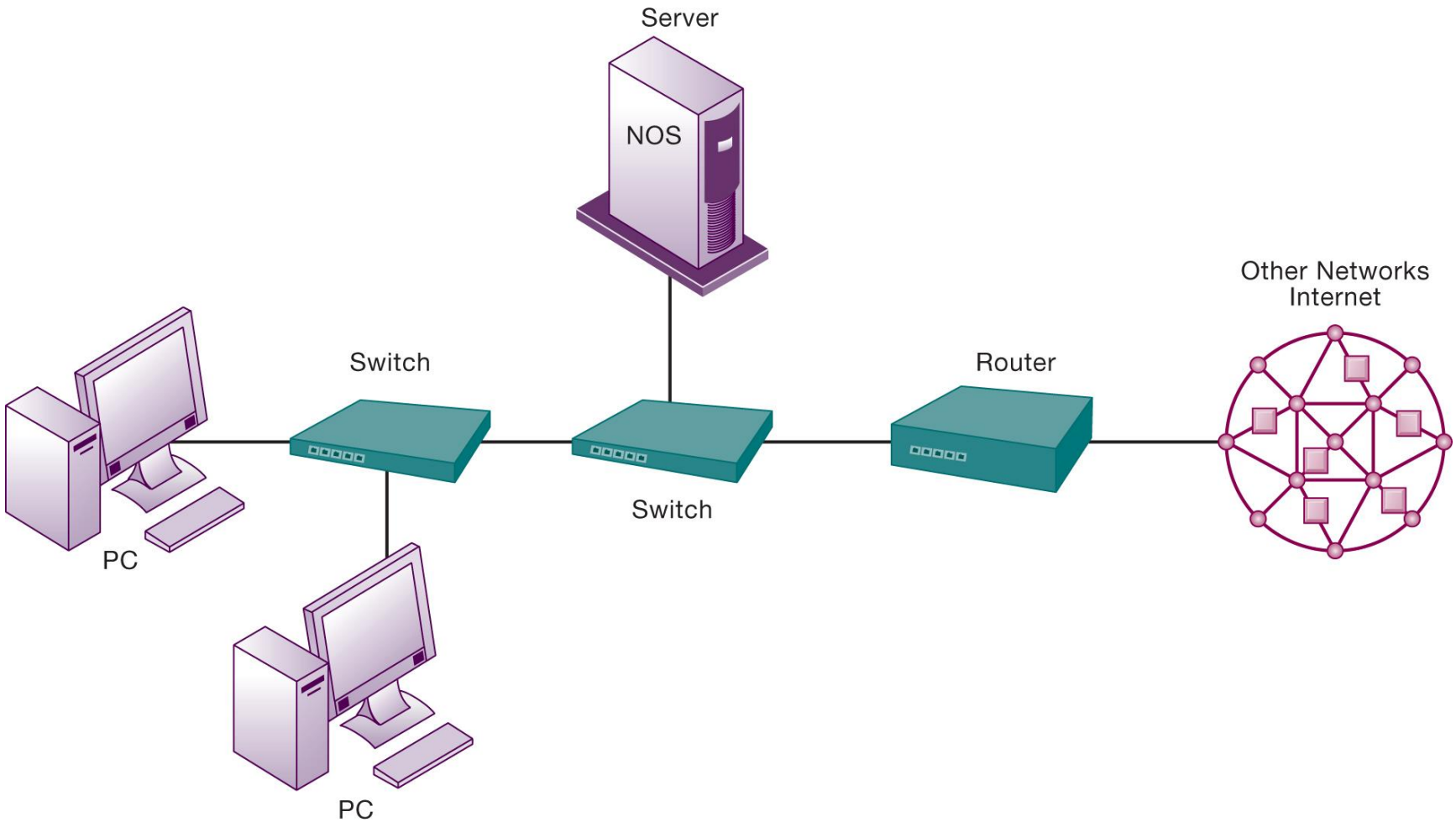
Networking and Communication Trends

- **Firms in the past used two different types of networks: *telephone networks* for voice communication and *computer networks* for data traffic.**
  - Computer networks have grown to include services like Internet telephony and video, enhancing their capabilities.
  - Voice and data communication networks have become faster, more mobile, and cheaper over time.
  - Merging into a unified digital network using Internet-based standards

- **A significant shift has occurred towards *broadband wireless platforms*, with over 70% of Internet users in the U.S. accessing the Internet via smartphones and tablets, in addition to desktop PCs.**

What Is a Computer Network?

- **Two or more connected computers**
- **Components of a Simple Computer Network**



What Is a Computer Network?

- **Network Interface Device (NID):** Connects computers to a network
- **Connection Medium**
  - Telephone wire, coaxial cable, fiber-optic cable, or radio signals for cell phones and Wi-Fi.
- **Network Operating System (NOS):**
  - ***Manages communications and coordinates resources***
  - Can be on every computer or a dedicated server
  - Examples: Microsoft Windows Server, Linux, Novell Open Enterprise Server
- **Server:**
  - Performs ***key functions for client computers***, such as displaying web pages, storing data, and hosting the network operating system (hence controlling the network).

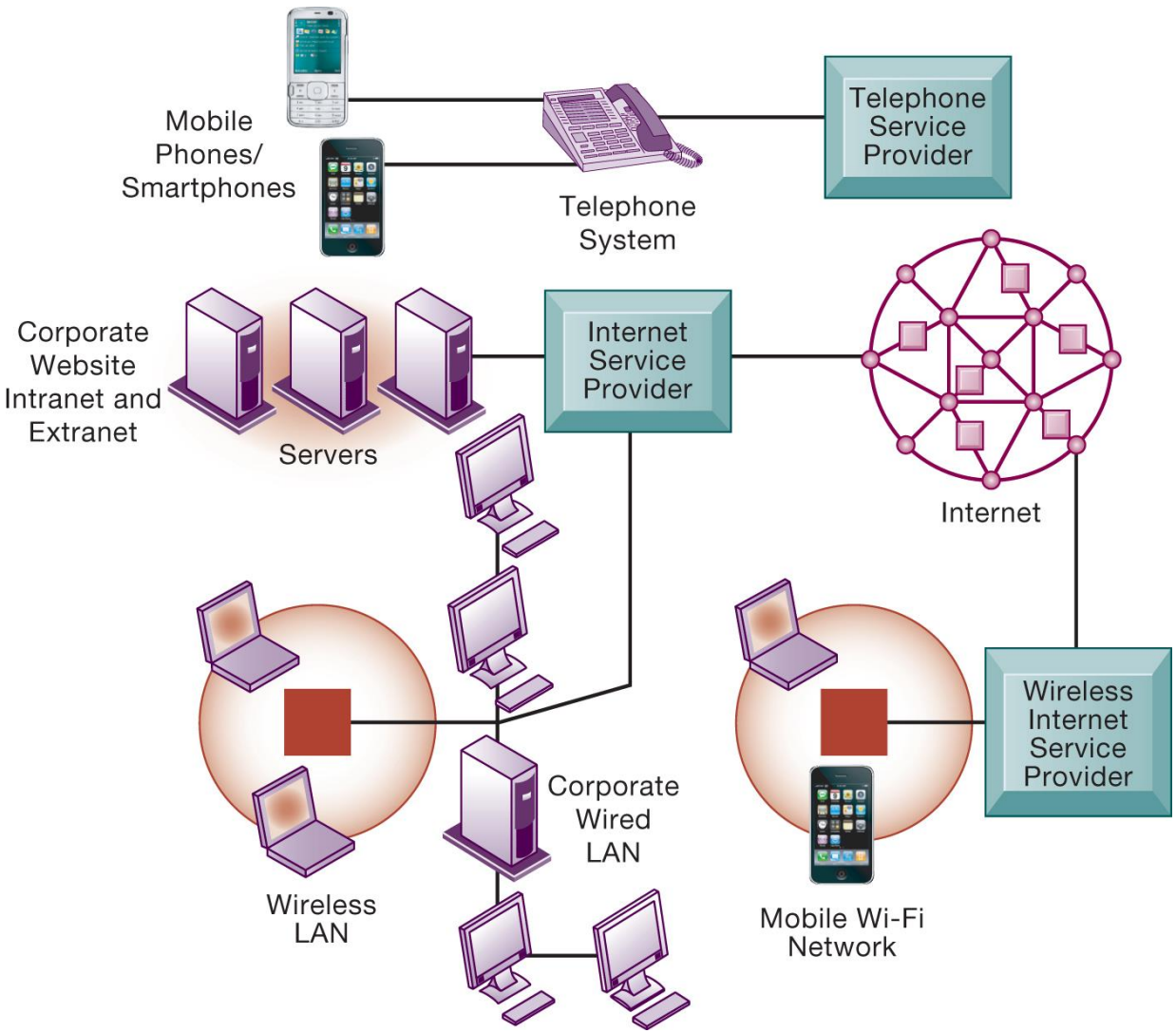
What Is a Computer Network?

- **Hubs and Switches**
  - Hubs ***connect*** devices and ***send*** data to all devices.
  - Switches are more advanced, sending data to ***specific destinations***.
- **Router**
  - Used to communicate with ***another network***, such as the Internet
  - Directs data packets through networks to the correct addresses.
- **Network Software**
  - Proprietary software in switches and routers directs data movement.
  - Can lead to bottlenecks and complexity in configuration.
- **Software-Defined Networking (SDN):**
  - Centralizes control functions via software.
  - Allows for ***efficient traffic management***, beneficial in cloud computing environments.

Networks in Large Companies

- **Various powerful servers and a lot of clients**
  - Powerful servers facilitate a corporate website, intranet, and possibly an extranet.
  - Servers are also connected to large computers for back-end system support.
- **Corporate Network Infrastructure:**
  - Consists of numerous ***interconnected local area networks (LANs)*** and firm-wide corporate networks.
    - Internet Service Provider
    - Telephone Service Provider
    - Wireless Internet Service Provider
- **Integration challenges:**
  - The complexity of integrating diverse communication networks and channels ***into a unified system*** for seamless information distribution within the company

Corporate Network Infrastructure



Key Digital Networking Technologies

- **Foundations of Digital Networks and the Internet:**
  1. Client/server computing
  2. Packet switching
  3. Communications standards like TCP/IP

Client/server Computing

- **Distributed computing model in which some of the processing power is located within *small, inexpensive* client computers**
  - Clients linked through network controlled by network server computer
  - Server sets rules of communication for network and provides every client with an address so others can find it on the network

- **Transition to Client/Server Computing**
  - ***Replaced the older model*** of centralized mainframe computing by extending computing capabilities to departments, workgroups, factory floors, etc.

- ***Personal Device* Connectivity:**
  - Client/server computing enables PCs, laptops, and cell phones to connect to networks like the Internet

Packet Switching

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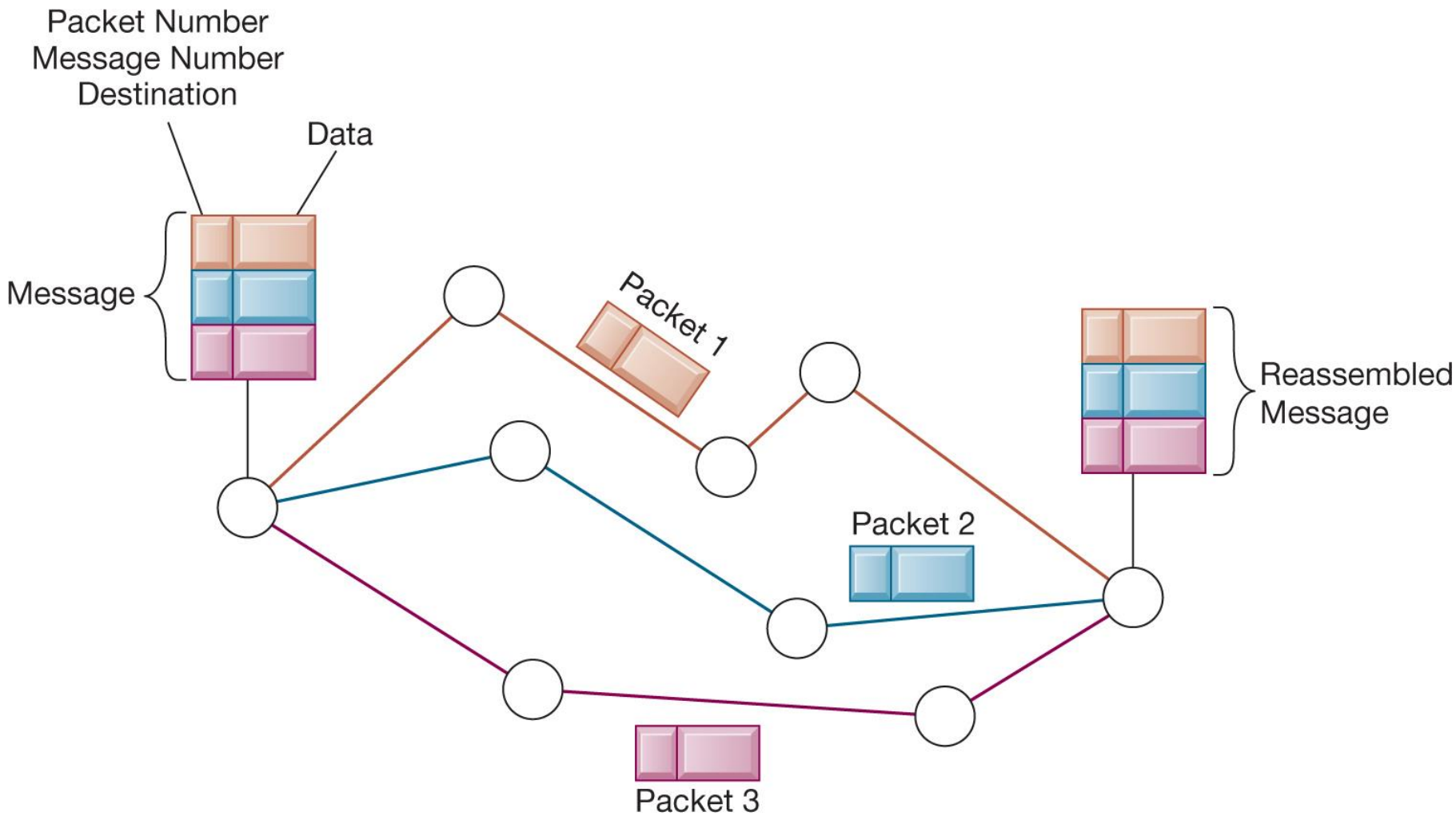
- **Packet Switching Basics:**
  - Digital messages are ***divided into packets***.
  - Packets are sent along various paths and ***reassembled*** at the destination.
- **Pre-Packet Switching Era: *Circuit-Switched* Networks**
  - Establish a direct and constant circuit between points for communication
  - Inefficient as the circuit remains maintained even when no data are sent

  ➔ Packing switching utilizes network capacity more effectively than circuit switching
- **How Packet Switching Works:**
  - Messages are ***split*** into packets with destination information and error-checking data.
  - Packets ***travel*** independently over different routes using routers.
  - Data packets from one source take many paths and are ***recombined*** at the destination.

Packet-Switched Networks and Packet Communications



TCP/IP

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- **TCP (Transmission Control Protocol) and IP (Internet Protocol) were developed to link different computers across heterogeneous networks.**
- **Telecommunication networks rely on diverse hardware and software, making *common protocols* essential for interoperability and reliable communication.**
- **Protocols define *shared rules* for how information is transmitted between network points**
  - Early proprietary protocols caused vendor lock-in, while
  - TCP/IP emerged as a ***widely adopted global standard***.
- **The TCP/IP protocol suite divides responsibilities:**
  - TCP establishes connections, sequences packets, and ensures reliable data transfer through acknowledgments.
  - IP handles packet addressing and routing, including packet disassembly and reassembly during transmission.

TCP/IP Model Layers and Others

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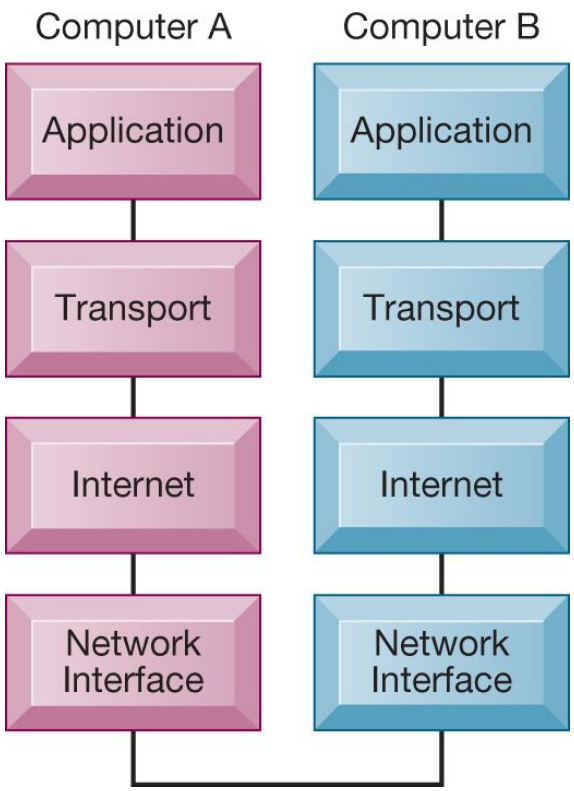
- **TCP/IP Model Layers:**
  - ***Application*** Layer: enables client application programs to access the other layers and defines the protocols (e.g. HTTP) that applications use to exchange data.
  - ***Transport*** Layer: providing the Application layer with communication and packet services
  - ***Internet*** Layer: addressing, routing, and packaging data packets called IP datagrams
  - ***Network Interface*** Layer: Interacts with the network medium for sending/receiving packets.
- **Cross-Platform Communication:**
  - Allows communication across different HW&SW platforms.
- **Data Transmission Process:**
  - Data moves down through the layers of the sending computer, across the network, and up through the layers of the receiving computer.
  - Damaged packets are identified and retransmitted upon request.



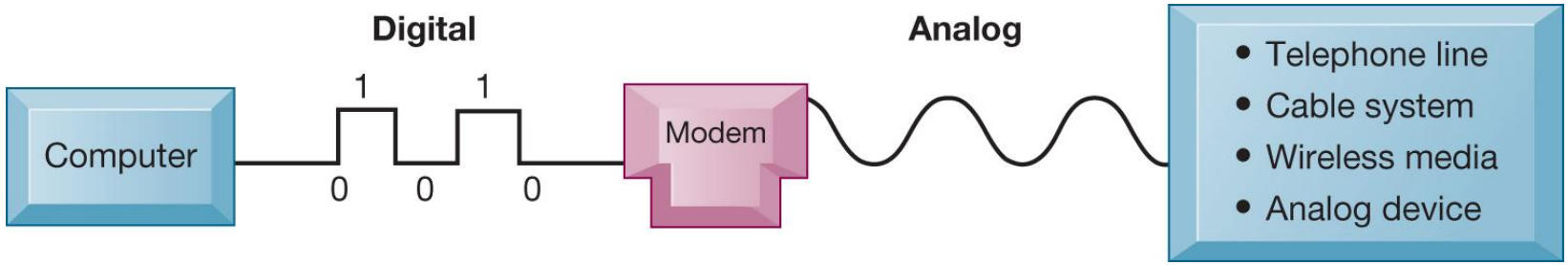
The Transmission Control Protocol /Internet Protocol (TCP/IP) Reference Model

7.2

What are the different types of networks?

Signals: Digital vs. Analog

- **An analog signal** is represented by a ***continuous waveform*** that passes through a communications medium and has been used for ***voice*** communication.
- **A digital signal** is a ***discrete, binary waveform*** rather than a continuous waveform.
- **Modem (modulator-demodulator) :** ***translates*** digital signals into analog form (and vice versa)
  - Cable modems, DSL modems, Wireless modems



Functions of the Modem

Types of Networks

1. **Local area networks (LANs)**

2. **Metropolitan Area Networks (MANs)**

3. **Wide Area Networks (WANs)**

Local area networks (LANs)

- **Designed to connect PCs and digital devices *within a 500m radius;* Commonly used in small offices, single or multiple buildings.**
  - Facilitate connections to WANs and other global networks via the Internet.
  - Typically includes ***a server*** that provides shared access to resources like software and data files; Incorporates ***a router*** to connect the LAN to external networks such as the Internet or other corporate networks.
  - Common LAN operating systems include Windows, Linux, and Novell.
- **Ethernet Standard**
  - ***Dominant LAN standard*** at the physical network level.
  - Specifies physical mediums, access control rules, and standardized data bits.
  - Originally supported 10 Mbps data transfer rates
- **LAN Architectures: client/server architecture vs. Peer-to-peer architecture (up to 10 users)**

Metropolitan and Wide Area Networks

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- **Metropolitan Area Networks (MANs)** span ***cities and major suburbs***, bridging the scale between WANs and LANs.
- **Wide Area Networks (WANs)** cover broad geographic areas such as ***regions or continents***, with the Internet being the most extensive WAN.
  - Connections to WANs are made via public networks, leased lines, or satellites.

TYPE	AREA
Local area network (LAN)	Up to 500 meters (half a mile); an office or floor of a building
Campus area network (CAN)	Up to 1,000 meters (a mile); a college campus or corporate facility
Metropolitan area network (MAN)	A city or metropolitan area
Wide area network (WAN)	A regional, transcontinental, or global area

Transmission Media Types and Speed

- **Networks utilize various physical transmission media:**
  - Twisted pair wire, coaxial cable, fiber-optic cable, and wireless transmission media.
- **Transmission speed** varies widely among different media, influenced by the specific HW & SW configurations used.

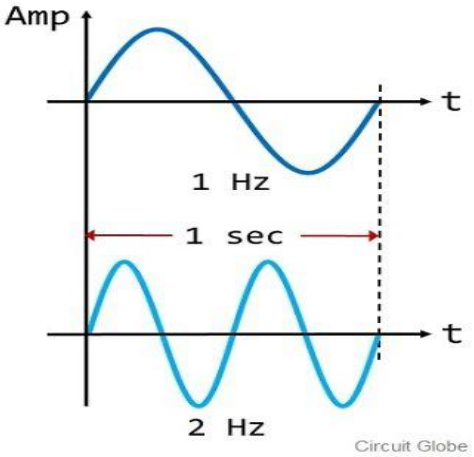
TRANSMISSION MEDIUM	DESCRIPTION	SPEED
Twisted pair wire	Strands of copper wire twisted in pairs for voice and data communications. Includes CAT5 and CAT6 cabling, with CAT6 capable of gigabit speeds.	CAT5: 10 Mbps-100 Mbps CAT6: Up to 10 Gbps
Coaxial cable	Thickly insulated copper wire, which is capable of high-speed data transmission and less subject to interference than twisted wire. Currently used for cable TV and for networks with longer runs (more than 100 meters).	Up to 1 Gbps
Fiber-optic cable	Strands of clear glass fiber, transmitting data as pulses of light generated by lasers. Useful for high-speed transmission of large quantities of data. More expensive than other physical transmission media; used for last-mile delivery to customers and the Internet backbone.	15 Mbps to 6+ Tbps
Wireless transmission media	Based on radio signals of various frequencies and includes both terrestrial and satellite microwave systems and cellular networks. Used for long-distance, wireless communication, and Internet access.	Up to 600+ Mbps

Transmission Media and Transmission Speed

TRANSMISSION MEDIUM	DESCRIPTION	SPEED
Twisted pair wire (CAT 5)	Strands of copper wire twisted in pairs for voice and data communications. CAT 5 is the most common 10 Mbps LAN cable. Maximum recommended run of 100 meters.	10–100+ Mbps
Coaxial cable	Thickly insulated copper wire, which is capable of high-speed data transmission and less subject to interference than twisted wire. Currently used for cable TV and for networks with longer runs (more than 100 meters).	Up to 1 Gbps
Fiber-optic cable	Strands of clear glass fiber, transmitting data as pulses of light generated by lasers. Useful for high-speed transmission of large quantities of data. More expensive than other physical transmission media and harder to install; often used for network backbone.	15 Mbps to 6+ Tbps
Wireless transmission media	Based on radio signals of various frequencies and includes both terrestrial and satellite microwave systems and cellular networks. Used for long-distance, wireless communication and Internet access.	Up to 600+ Mbps

Bandwidth and Transmission Capacity

- **Bandwidth refers to the *total capacity* of a telecommunications medium to transmit digital information, measured in bits per second (bps).**
- **The transmission capacity of a medium is *directly linked to its frequency*, measured in hertz (Hz)**
  - Bandwidth is determined *by the range of frequencies* a medium can support
- ***A wider frequency range results in higher bandwidth*, enhancing the medium's capacity for data transmission.**



7.3

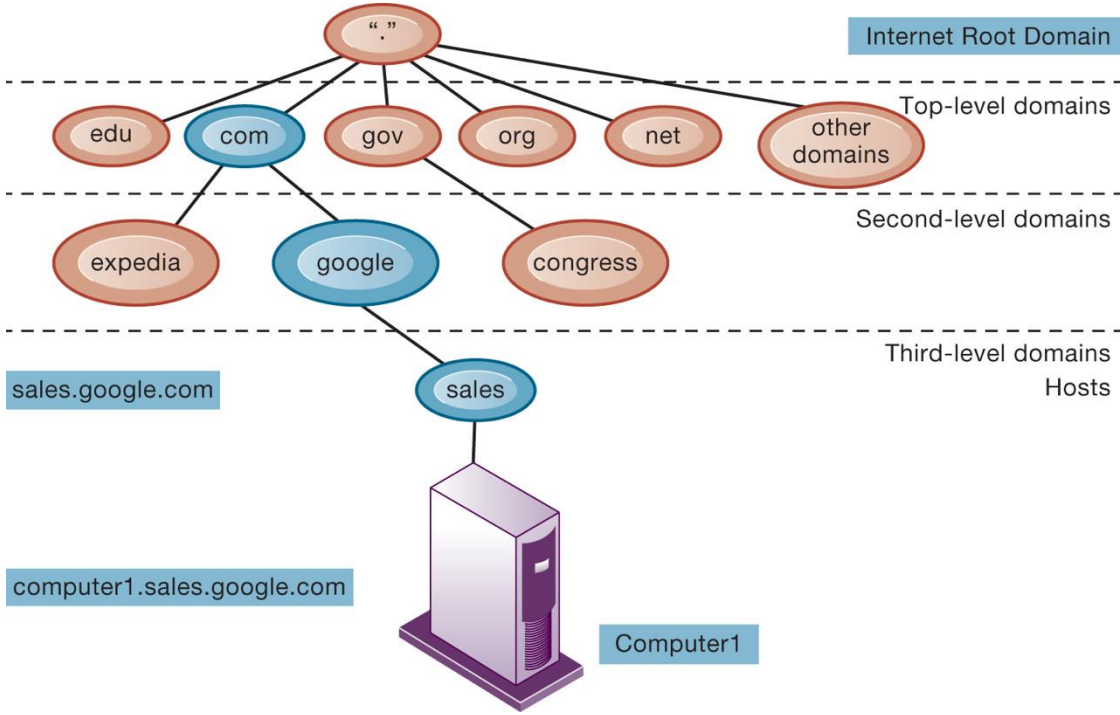
How do the Internet and Internet technology work, and how do they support communication and e-business?

What Is the Internet?

- **World's most extensive network**
- **Internet service providers (ISPs)**
  - A commercial organization with a permanent connection to the Internet
    - Example: EarthLink, NetZero, AT&T, KT, U+
  - Universities, research centers, and businesses often have ***their own designated*** Internet domains for connectivity.
  - Types of Internet Connections
    - Digital subscriber line (DSL/FIOS): existing telephone lines, 385Kbps–40Mbps
    - Cable Internet connections: digital cable coaxial lines, 1–50 Mbps
    - Satellite: generally slower upload speeds
    - Dedicated lines like T1 (1.54 Mbps) and T3 (45 Mbps) for businesses requiring high-speed and reliable service.

Internet Addressing and Architecture

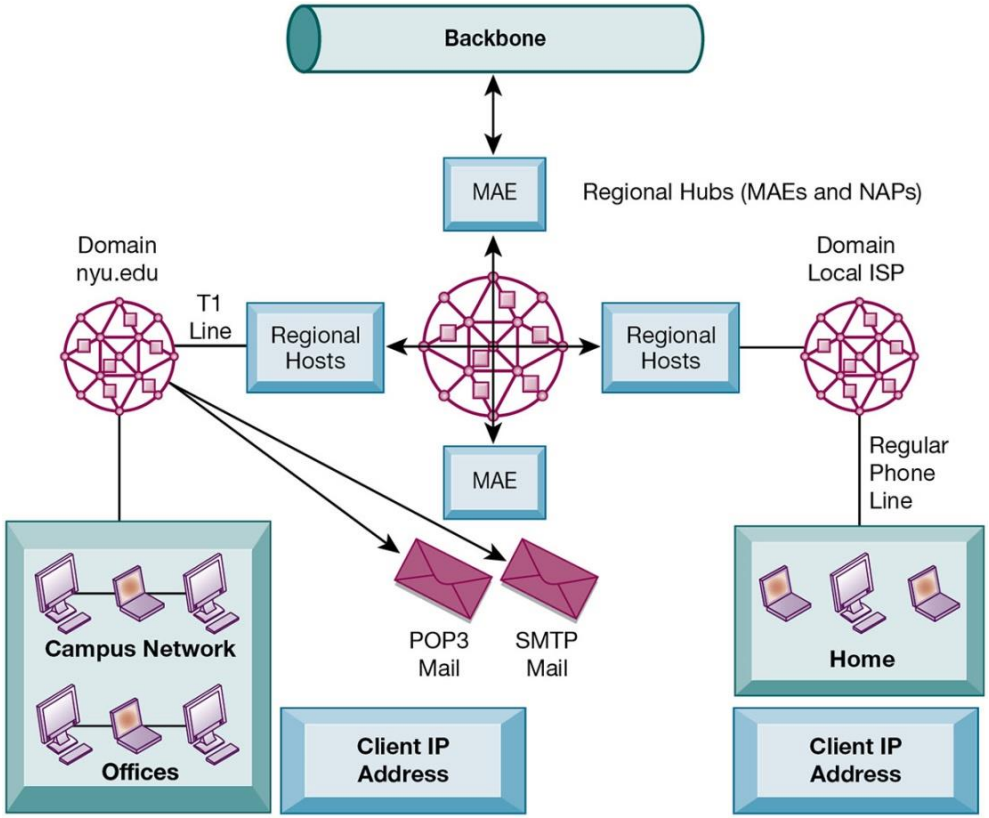
- **Each device on Internet assigned IP address**
  - 32-bit number
  - e.g. 207.46.250.119, 141.223.XXX.XXX
- **Domain Name System (DNS)**
  - Converts IP addresses to domain names
  - Hierarchical structure
  - Top-level domains
    - .com, .edu, .gov, .org
  - Second-level:
    - name of a company/organization
  - Third-level:
    - a specific computer/service



The Domain Name System

Internet Architecture and Governance

- **Network service providers**
  - Own trunk lines (high-speed backbone networks)
- **Regional telephone and cable TV companies**
  - Provide regional and local access
- **Professional organizations and government bodies establish Internet standards**
  - IAB, ICANN, W3C



The Future Internet: IPv6 and Internet 2

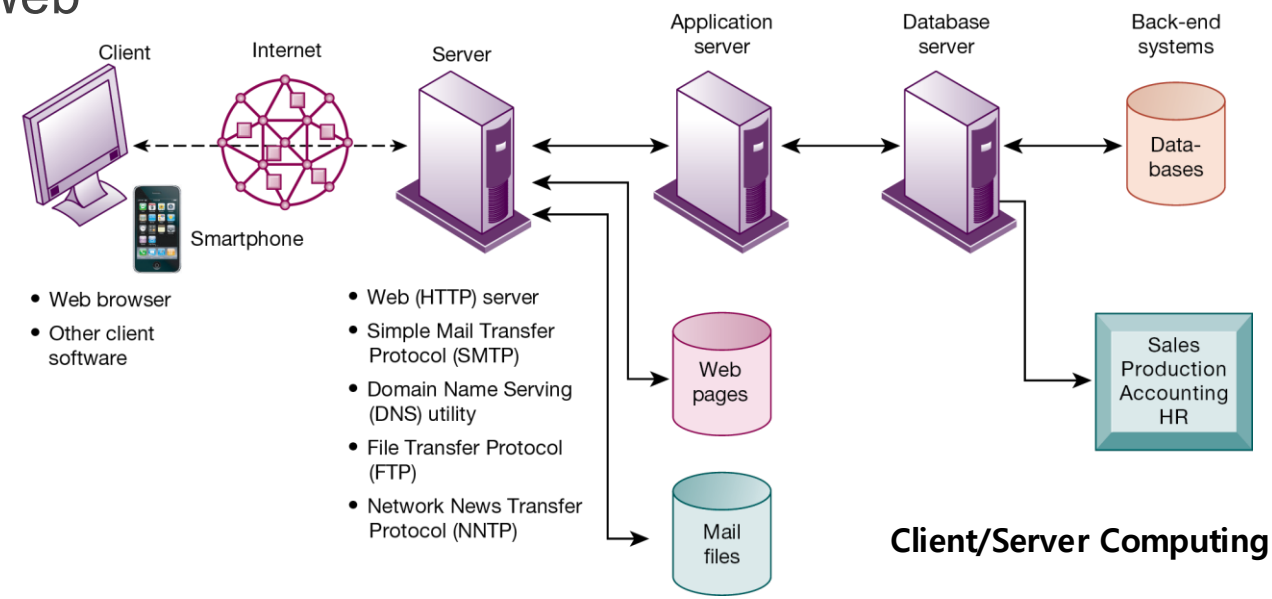
- **IPv6 (Internet Protocol version 6)**
  - Running out of available IP addresses using the old addressing convention
  - ***New addressing scheme*** for IP numbers
  - Will provide more than a quadrillion new addresses, contains 128-bit addresses (2 to the power of 128)
  - ***Not compatible*** with earlier IPv4 addressing

- **Internet2**
  - ***Advanced networking consortium***
    - Universities, businesses, government agencies, other institutions
  - Developed high-capacity 100 Gbps testing network
  - Testing leading-edge new technologies for Internet

Internet Services and Communication Tools

- **Internet services**
  - E-mail
  - Chatting and instant messaging
  - Newsgroups
  - Telnet
  - File Transfer Protocol (FTP)
  - World Wide Web

CAPABILITY	FUNCTIONS SUPPORTED
Email	Person-to-person messaging; document sharing
Chatting and instant messaging	Interactive conversations
Newsgroups	Discussion groups on electronic bulletin boards
Telnet	Logging on to one computer system and doing work on another
File Transfer Protocol (FTP)	Transferring files from computer to computer
World Wide Web	Retrieving, formatting, and displaying information (including text, audio, graphics, and video) by using hypertext links



Client/Server Computing on the Internet

FileZilla

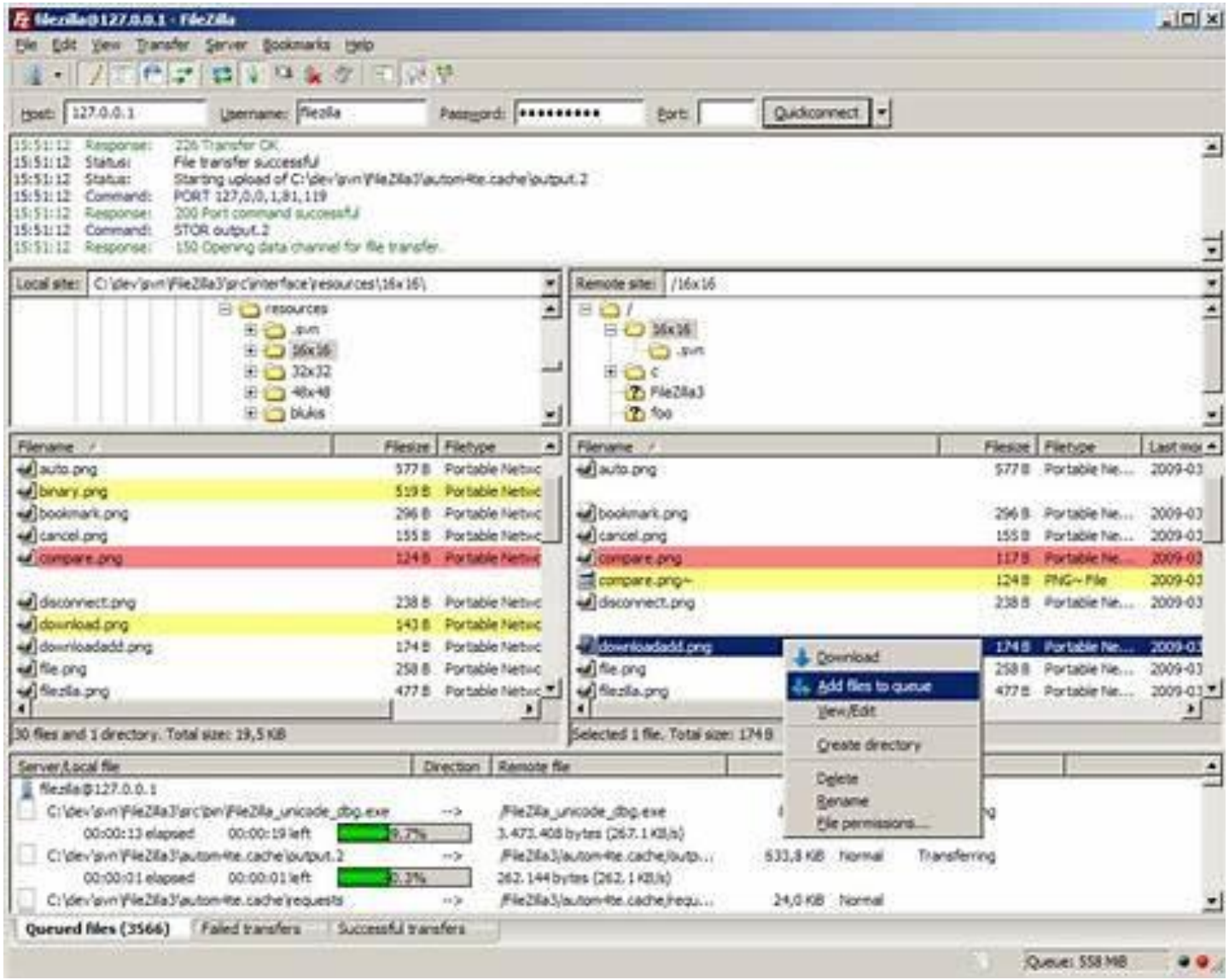
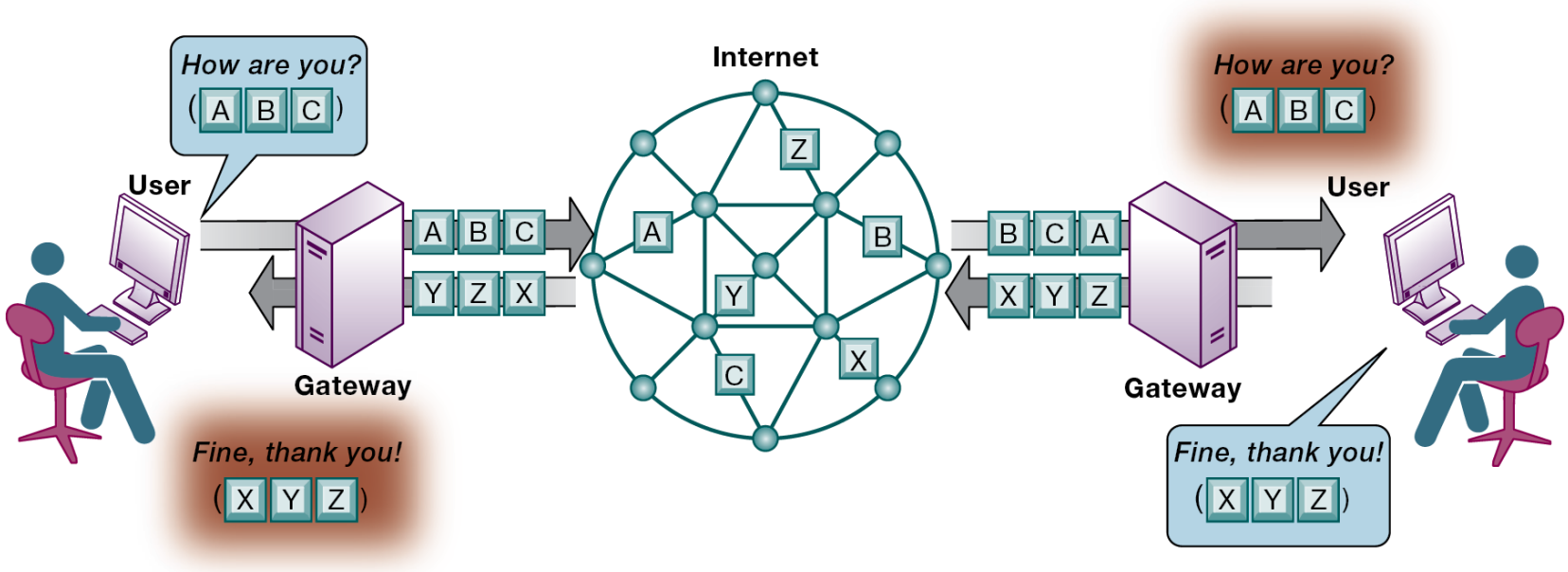


Image Source: https://www.blogdumoderateur.com/tools/filezilla/

Internet Services and Communication Tools: VoIP

- **Voice over IP (VoIP)**
  - Digital voice communication using IP, packet switching
  - Avoid the tolls charged by local and long-distance telephone networks
    - VoIP can ***reduce*** communication and network management ***costs*** by 20~30 percent.



Internet Services and Communication Tools: Other

- **Unified communications**
  - Communications systems that integrate voice, data, e-mail, conferencing
- **Virtual private network (VPN)**
  - Secure, encrypted, private network run over the Internet
  - PPTP (Point-to-Point Tunneling Protocol)
  - ***Tunneling***: packets of data are encrypted and wrapped inside IP packets.

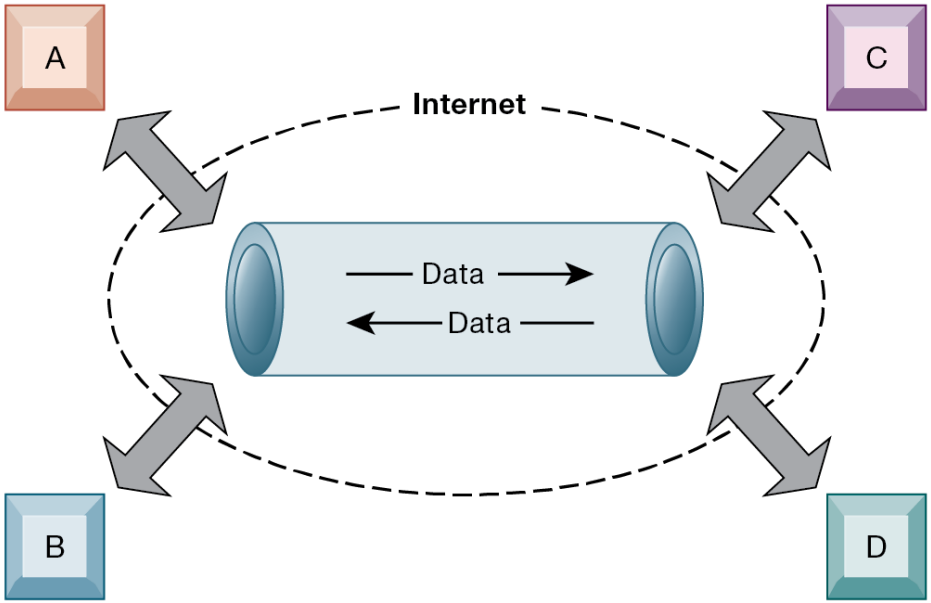


Image Source: tunnelbear.com

A Virtual Private Network Using the Internet

The Web, Hypertext, and Web servers

- **The Web**
  - A system with ***universally accepted standards*** for storing, retrieving, formatting, and displaying information by using a ***client/server architecture***
  - Web pages are formatted using ***hypertext*** with embedded links that connect documents to one another and that also link pages to other objects
- **Hypertext**
  - Hypertext Markup Language (HTML): formats documents and incorporates dynamic links to other docs and pictures stored in the same or remote computers
  - Hypertext Transfer Protocol (HTTP): communications standard that transfers pages on the web.
  - Uniform resource locator (URL): where to look for the information
- **Web servers**
  - Software for ***locating and managing*** web pages
  - Apache HTTP Server, Microsoft Internet Information Services (IIS)

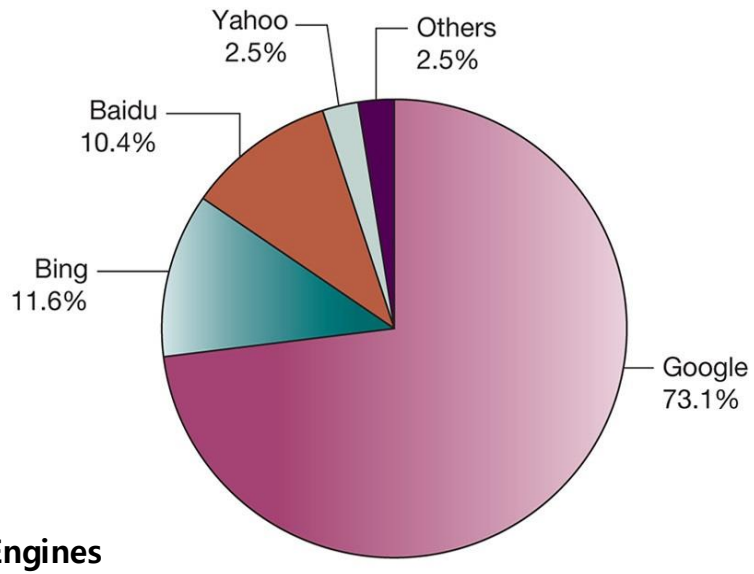
Searching for Information on the Web

- **Introduction to Web Searching**
  - ***Search engines*** serve as major access points for finding information on the web
    - As of 2011, Google indexed over 50B, representing only a portion of the web

- **Deep Web and Closed Web Systems**
  - The 'deep web' is not indexed by standard search engines.
  - It's estimated to contain around ***1 trillion pages***, including subscription sites like Wall Street Journal Online and protected corporate databases.
  - Platforms like Facebook represent ***a closed web system*** where content is ***not fully searchable*** by external search engines.
  - The ***dark web***, a smaller segment of the deep web, is intentionally hidden and used for anonymous transactions, often illegal.

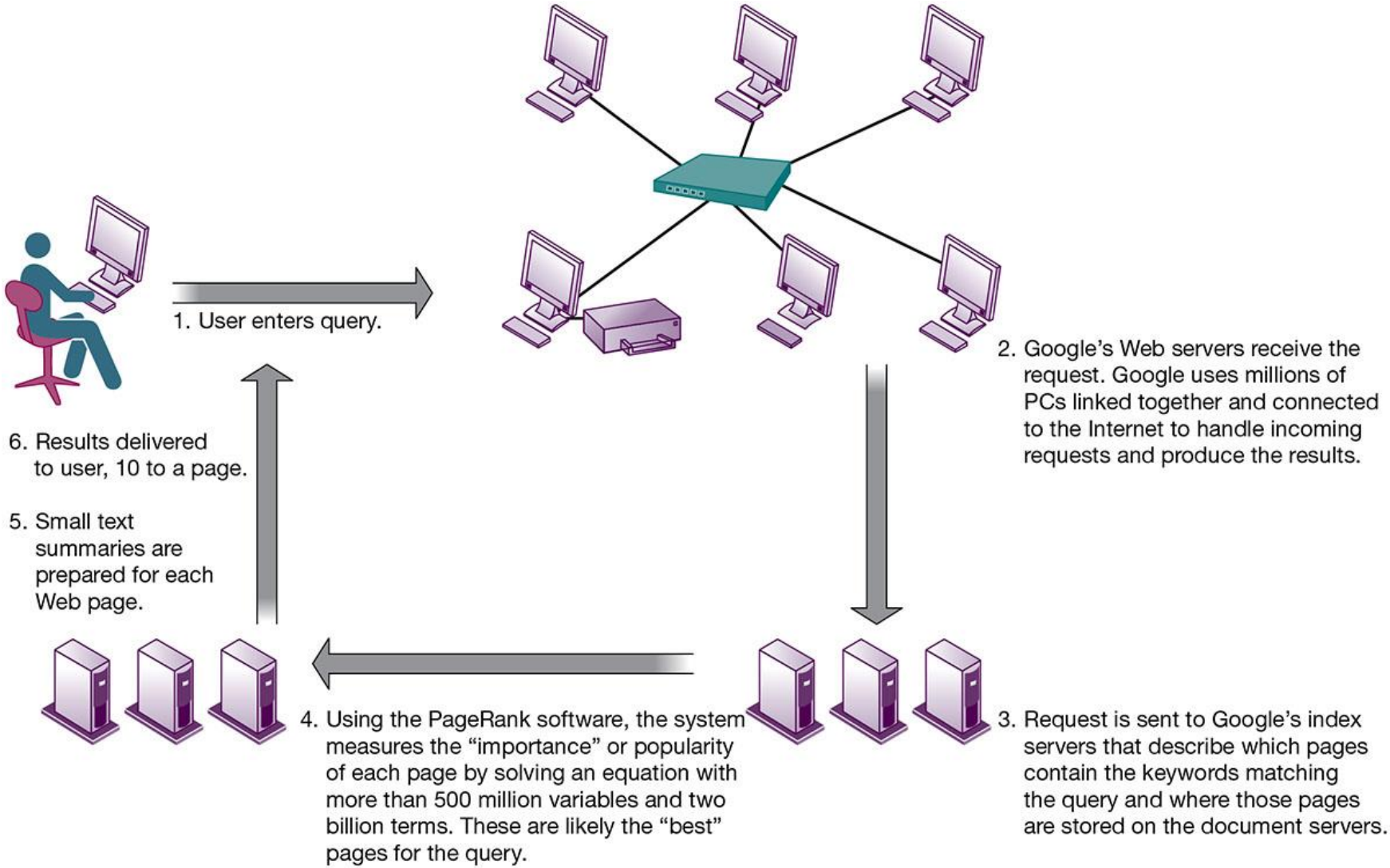
Function and Evolution of Search Engines

- **Google introduced ranking based on page popularity and complex indexing techniques.**
  - Google's ***PageRank*** System: essentially measures the popularity of a web page by calculating the number of sites that link to that page as well as the number of pages to which it links
  - Unique web crawler program that indexed not only keywords on a page but also ***combinations of words*** (such as authors and the titles of their articles)



Top U.S. Web Search Engines

How Google Works



Mobile Search & Semantic Search

- **Mobile Search**
  - Smartphone and tablet searches currently constitute over ***50% of all*** internet searches.
  - Companies like Google, Amazon, and Yahoo have improved their mobile search interfaces to enhance ease.

- **Semantic Search**
  - Semantic search aims to improve search engines by ***making them understand human language and intent*** more accurately.
  - Google's Hummingbird algorithm was introduced to enhance semantic search
  - Google searches also take advantage of ***Knowledge Graph***
  - Google has made ***predictive search*** part of most search results

Visual Search, Visual Web, and Visual Social Networks

- **Visual search and the visual web**
  - The explosion of ***photos and videos*** on the Internet created a demand for searching and classifying these visual objects.
    - Facebook uses AI to enhance its facial recognition technology, allowing users to search for and identify people based on their digital images.
  - The ***visual web*** refers to websites such as Pinterest, where pictures replace text documents, users search for pictures, and pictures of products replace display ads for products.

- **Visual Social Networks**
  - ***Pinterest*** is a major player in the visual web, functioning as a social networking site where users and brands create boards for pinning images
  - ***Instagram*** is another visual web, serving as a photo and video sharing platform

Search engine marketing

- **Search engines have become *major advertising platforms and shopping tools* by offering what is now called search engine marketing.**

- **Over *90 percent of web traffic* originates from search engines, with search engine marketing constituting about 44 percent of digital advertising expenditure.**

- **Types of Search Listings**
  - Search results typically feature two types of listings: ***sponsored*** links (paid advertisements) and ***unsponsored, organic*** search results.

Effectiveness and Techniques of Search Engine Marketing

- **Search engine marketing is highly effective, boasting the highest *click-through rates* and returns on advertising investments.**

- ***Search Engine Optimization (SEO)* is employed to improve a website's visibility and ranking on search engines**
  - Using techniques such as aligning web content with likely search terms (e.g., "lighting" vs. "lamps") and increasing links from other sites.

- **SEO practices enhance a site’s ranking by making it more relevant and popular according to search engine algorithms.**

Challenges and Benefits of SEM

- **Challenges and Manipulations**
  - Search engines face challenges from ***scammers*** who create fake pages or buy links from "link farms" to artificially boost site rankings.
  - Google and other search engines continuously ***update their algorithms*** to prioritize the quality of links and penalize suspicious linking patterns.

- **Benefits to Small Businesses and Consumers**
  - Search engines ***level the playing field*** for small businesses by allowing them to reach "hot prospects"
  - Advertisers pay only for ads that ***generate click-throughs***, not for mere ad displays, making search engine marketing cost-effective.
  - Consumers benefit from seeing ads only for products they are ***actively seeking***, reducing their cognitive load and search costs.

Sharing Information on the Web

- **Blogs and microblogs:** a personal website that typically contains a series of chronological entries by its author and links to related web pages.
- **RSS:** pulls specified content from websites and feeds it automatically to users' computers
- **Wikis:** collaborative websites where visitors can add, delete, or modify content on the site
- **Social networking:** enable users to build communities of friends and share information

The Future Web

- **Pervasive web:** controls everything from a city's traffic lights and water usage, to the lights in your living room, to your car's rearview mirror
- **Internet of Things:** based on billions of Internet-connected sensors throughout our physical world
- **Internet of People:** based on ***sensors attached to clothing and personal effects*** that monitor physical states and locations of individuals
- **App Internet:** direct access to content
- **Increased cloud computing and SaaS**
- **Ubiquitous mobile connectivity**
- **Greater seamlessness of web as a whole**

7.4

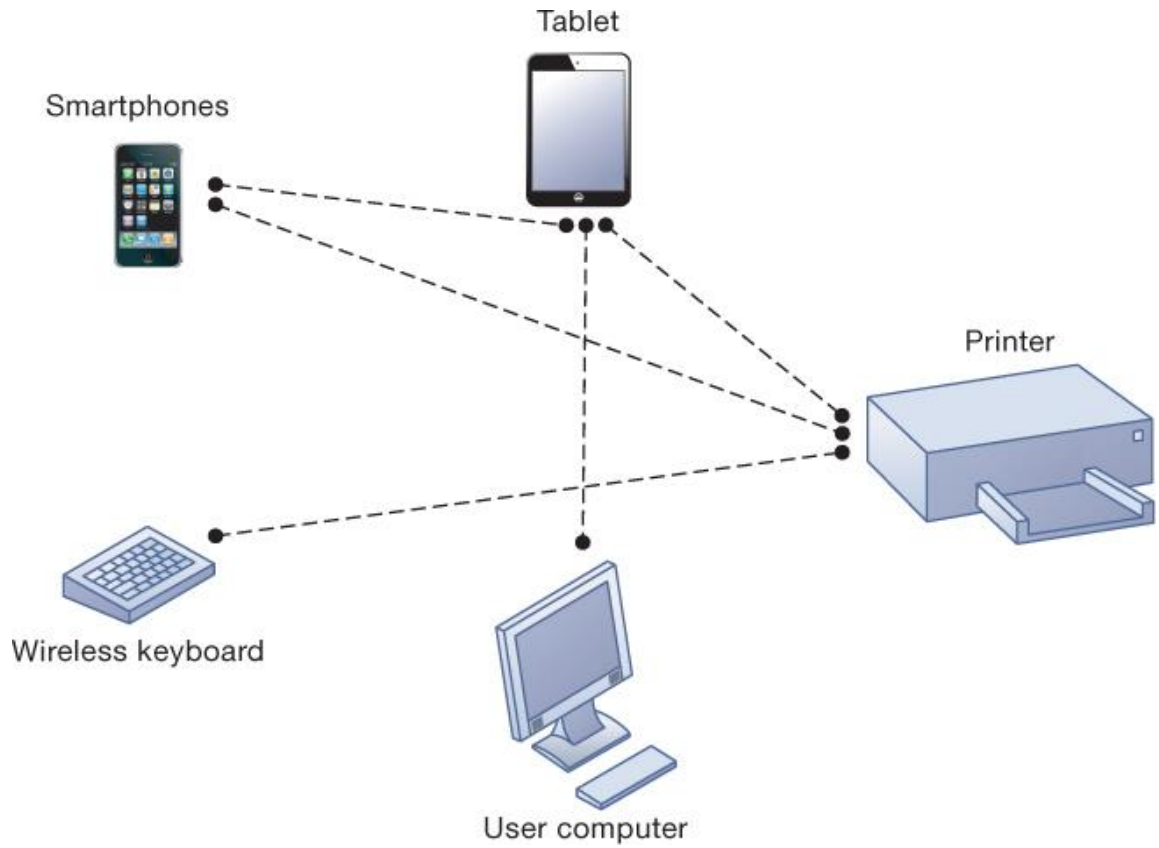
What are the principal technologies and standards for wireless networking, communication, and Internet access?

Cellular Systems

- **Competing standards**
  - CDMA: United States, Korea only
  - GSM: Rest of world, AT&T, T-Mobile
- **Third-generation (3G) networks**
  - 144 Kbps
  - Suitable for ***e-mail access, web browsing***
- **Fourth-generation (4G) networks**
  - Up to 100 Mbps
  - Suitable for ***Internet video***
  - LTE and WiMax
- **5G Networks**
  - Gigabit capacity

Wireless Computer Networks and Internet Access

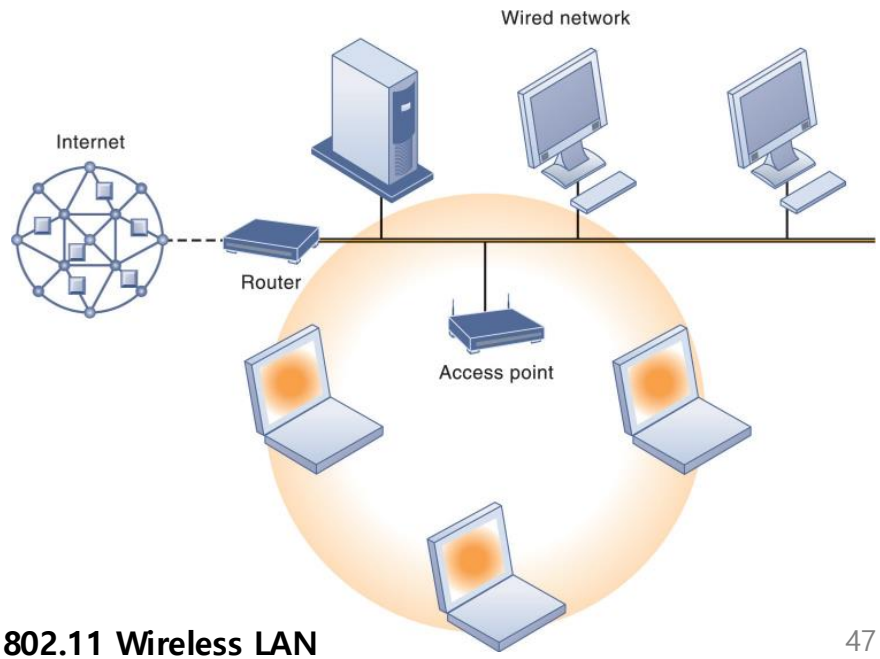
- **Bluetooth (802.15)**
  - Links ***up to 8 devices in 10-m area*** using low-power, radio-based communication
  - Useful for personal networking (PANs)



A Bluetooth Network (PAN)

Wi-Fi and WiMax

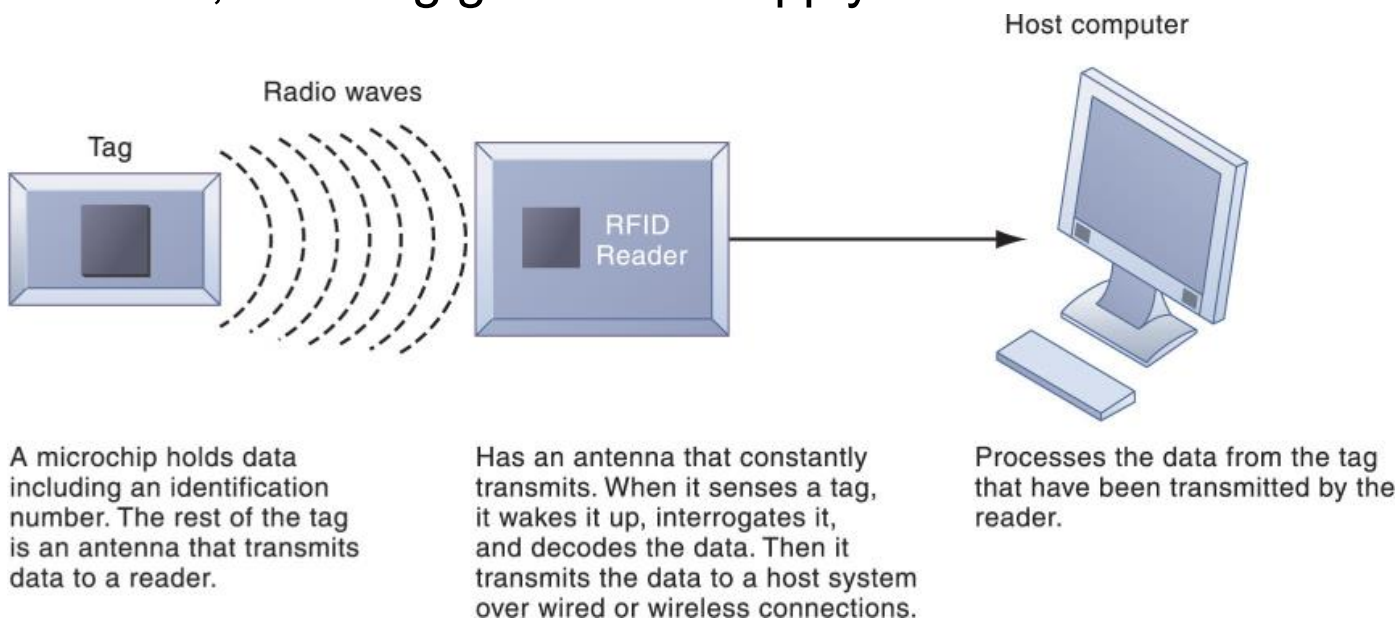
- **Wi-Fi (802.11)**
  - Used for ***wireless LAN and wireless Internet*** access
  - Effective distance of ***30 to 50 meters***
  - Use access points: device with radio receiver/transmitter for connecting wireless devices to a wired LAN
  - Hotspots: one or more access points in public place to provide maximum wireless coverage for a specific area
  - Weak security features
- **WiMax (802.16)**
  - Wireless access range of ***30 miles*** and transmission speed of up to ***75 Mbps***
  - Require WiMax antennas



An 802.11 Wireless LAN

Radio Frequency Identification (RFID)

- **Use tiny *tags with microchips* containing data about an item and location**
- **Tag antennas to transmit radio signals over short distances to special RFID readers**
- **Common uses:**
  - Automated toll-collection; Tracking goods in a supply chain



How RFID Works

Near Field Communication (NFC)

- **A short-range wireless connectivity standard that uses *electromagnetic radio fields* to enable two compatible devices to exchange data when brought within a few cm of each other**
  - Used in ***Tap-and-go services*** like Apple Pay or Google pay
  - A smartphone sends out radio frequency signals that interact with an NFC tag found in compatible card readers or smart posters.
  - The signals create a current that flows through the NFC tag, allowing the device and the tag to communicate with one another.
  - In most cases the tag is passive and only sends out information while the other device is active and can both send and receive information.
- **Used in *wireless payment* services to retrieve information**

Wireless Sensor Networks (WSNs)

- Networks of ***hundreds or thousands of*** interconnected wireless devices
- Used to ***monitor*** building security, ***detect*** hazardous substances in the air, monitor environmental changes, traffic, or military activity
- Devices have built-in processing, storage, and radio frequency sensors and antennas
- Require ***low-power, long-lasting batteries*** and the ability to endure in the field without maintenance
- Major sources of "Big Data" and fueling "Internet of Things."

Key Takeaways

- Computer networks connect devices to share data and resources, ranging from simple local networks to complex corporate infrastructures
- Modern networks are built on client/server computing, packet switching, and TCP/IP
- Different network types serve different geographic scopes, including LANs (local), MANs (city-wide), and WANs (regional or global)
- The Internet is a global network of networks, identified by IP addresses and supported by DNS, standards organizations, and core services such as email, the web, file transfer, and messaging
- Wireless, mobile, and emerging technologies are expanding network capabilities, including cellular networks (3G/4G), Wi-Fi, Bluetooth, VoIP, VPNs, RFID, and wireless sensor networks